BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 069 · Functions and Return Values

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a record?

Q2. Why retest a program after a fix?
A. Check for new faults
B. Prove no faults can exist
C. Avoid testing old features
D. Remove all input rules

Q3. “The Internet is a single computer containing every website.” Correct this. Explain why.

Q4. A browser asks for a school's web page. “A client must host a website before asking for it.” Correct this. Which computer provides the page?

I Do · Read new knowledge

Key vocabulary

Function — Named reusable subprogram that returns a value to its caller.

Parameters — Named inputs that allow a function to work on supplied arguments.

Return — A RETURN statement sends a calculated value to the calling code.

Modularity — Dividing a program into functions makes sections easier to reuse and test.

Algorithm — A defined set of processing instructions.

Testing — Checking expected outcomes against actual program behaviour.

New learning

Function: Named reusable subprogram that returns a value to its caller. In the rectangle-area calculator, identify where this technique applies.

Parameters: Named inputs that allow a function to work on supplied arguments. In the taxi fare function, identify where this technique applies.

Return: A RETURN statement sends a calculated value to the calling code. In the test-mark average, identify where this technique applies.

Modularity: Dividing a program into functions makes sections easier to reuse and test. In the triangle area, identify where this technique applies.

Worked example

A rectangle-area calculator uses function and parameters. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Function [1]; Named reusable subprogram that returns a value to its caller. [1].

3. Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do · Practise with guidance

Guided 1

A taxi fare function needs function. Explain the purpose. [2 marks]

Worked method: Function [1]; Named reusable subprogram that returns a value to its caller. [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

test. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Named fields about one item or person. Fields may have different data types.

Q2. A — Check for new faults

Q3. A worldwide collection of interconnected computer networks.

Q4. A web server stores website resources and sends them when clients request them. Access needs working servers and network services.

I Do / We Do pages · Guided practice

Q1. Function [1]; Named reusable subprogram that returns a value to its caller. [1].

Q2. Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].

Q3. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and test. [1].

You Do pages · Code First

```
function pixels(width, height)
    return width * height
endfunction
print(pixels(4, 3))
```

Expected result: 12

You Do pages · Independent questions

Q1. Function [1]; Parameters [1].

Q2. Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1]. Return [1]; A RETURN statement sends a calculated value to the calling code. [1].

Q3. Function [1]; Named reusable subprogram that returns a value to its caller. [1]. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Guided 2

Explain parameters for a test-mark average. [2 marks]

Worked method: Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].

Guided 3

When programming a triangle area, explain why return and modularity matter. [4 marks]

Worked method: Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and test. [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the return statement so the function calculates a pixel count.

```
function pixels(width, height)
    return ...
endfunction
print(pixels(4, 3))
```

write

Question 1

State TWO relevant ideas or operations from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain parameters and return in a shopping cost function. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

A programmer makes a class total calculator. Explain the roles of function, return and modularity. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A function is a named reusable subprogram that returns a value to its caller.